

The Origin of G_2 Holonomy and the Sedenion Scream: A Gravitational Wave Signature at 1.8 GHz

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Abstract

We propose a cosmological framework wherein the pre-geometric vacuum is identified with the full hierarchy of Cayley-Dickson algebras ($\mathbb{A}_n$). We demonstrate that the “Big Bang” corresponds to a topological filtration event driven by the distinct algebraic requirements of Time and Space. While the metric geometry of space collapses in dimensions $n \geq 16$ due to the proliferation of zero divisors ($xy = 0$), the temporal evolution remains unitary due to the universal property of *Power Associativity* ($x^n x^m = x^{n+m}$). We identify the physical vacuum as the stable residue of this filtration. Utilizing recent results by Reggiani (2024), we prove that the manifold of excluded Sedenion zero divisors $\mathcal{Z}(\mathbb{S})$ is isometric to the exceptional Lie group G_2 . Consequently, the observed compactification of M-theory on a manifold of G_2 holonomy is derived as the boundary condition of the algebraic void left by the expulsion of the zero-divisor swarm. Finally, using the scale-invariant Thin Filament Code (TFC) solver with a critical stiffness of $\Gamma = \pi$, we predict the fundamental resonance frequency of the Sedenion transition to be $\omega_{\mathbb{S}} \approx 6.81$ (dimensionless units). Scaling this to the GUT symmetry-breaking scale yields a distinct, narrow-band gravitational wave signal at **1.8 GHz**. We outline an experimental search strategy for this signal using existing axion haloscopes via the Inverse Gertsenshtein Effect.

1 Introduction: The Algebraic Ecology of the Vacuum

Standard M-theory compactifications rely on manifolds of G_2 holonomy to preserve $\mathcal{N} = 1$ supersymmetry in four dimensions. However, the selection principle for this specific holonomy group remains an open question. We propose that G_2 geometry is not an arbitrary choice, but a necessary boundary condition imposed by the algebraic stability of the pre-geometric vacuum.

We postulate that the initial cosmological state is a **Plenum** containing the full tower of Cayley-Dickson algebras $\mathbb{A}_n$ (Real, Complex, Quaternion, Octonion, Sedenion, Trigintaduonion, ...). The dynamics of cosmogenesis are driven by the competition between two fundamental imperatives:

1. **Temporal Persistence:** The requirement for unitary time evolution $U(t)$.
2. **Spatial Observability:** The requirement for a non-degenerate metric norm $\|xy\| = \|x\|\|y\|$.

2 The Power-Associative Lifeline: The Persistence of Time

A critical distinction exists between the algebraic requirements for time versus space. This distinction permits the universe to evolve temporally even before spatial geometry is stabilized.

2.1 Universal Unitary Evolution

Time evolution is generated by the exponentiation of a Hamiltonian operator H . A fundamental property preserved throughout the entire Cayley-Dickson construction (even for the 32-dimensional Trigintaduonions $\mathbb{T}$ and beyond) is **Power Associativity**:

$$x^n \cdot x^m = x^{n+m}, \quad \forall x \in \mathbb{A}_n \quad (1)$$

This property ensures that the one-parameter group of time evolution operators is well-defined even in the high-dimensional bulk algebras:

$$U(t) = e^{-iHt} = \sum_{k=0}^{\infty} \frac{(-iHt)^k}{k!} \quad (2)$$

Because powers associate, the composition law $U(t_1)U(t_2) = U(t_1 + t_2)$ holds. Therefore, **Time is fundamental**. The causal clock ticks consistently regardless of the dimension or non-associativity of the substrate.

2.2 The Spatial Catastrophe

In contrast, spatial geometry relies on the existence of a division algebra to define a consistent metric and interaction vertex. For $n \geq 4$ (Sedenions $\mathbb{S}$ and beyond), the algebras lose the division property and admit **Zero Divisors**:

$$\exists u, v \in \mathbb{A}_n, u \neq 0, v \neq 0 \text{ such that } u \cdot v = 0 \quad (3)$$

Physically, a zero divisor represents a **Vacuum Crash**. If two non-zero states interact to produce zero, information is destroyed locally. The spatial metric becomes singular, and the norm property fails ($\|uv\| \neq \|u\|\|v\|$). Thus, while Time persists in the bulk, Space is dynamically unstable. The “Big Bang” is the phase transition where Space attempts to catch up to Time by purging these algebraic errors.

3 The Reggiani Bridge: Deriving G_2 from Failure

To satisfy the Axiom of Observability for spatial interactions, the vacuum must excise the zero-divisor regions. We focus on the first critical failure at $n = 4$ (Sedenions).

3.1 The Manifold of Zero Divisors

Let $\mathcal{Z}(\mathbb{S}) \subset \mathbb{S} \times \mathbb{S}$ be the manifold of normalized zero divisors in the Sedenion algebra. The physical vacuum $\mathcal{M}_{phys}$ is the complement:

$$\mathcal{M}_{phys} \cong \mathbb{S} \setminus \mathcal{Z}(\mathbb{S}) \quad (4)$$

The topology of the observable universe is determined by the boundary of this excluded hole.

3.2 Theorem: The Isometry to G_2

We utilize the rigorous result established by Reggiani (2024):

Theorem 1 (Reggiani): The manifold of zero divisors $\mathcal{Z}(\mathbb{S})$ is isometric to the compact exceptional Lie group G_2 equipped with a naturally reductive left-invariant metric.

$$\mathcal{Z}(\mathbb{S}) \cong G_2 \quad (5)$$

This establishes a direct link between algebraic failure and geometric structure. The G_2 manifold of M-theory is physically identified as the **Topological Scar** formed by the removal of Sedenionic zero divisors from the vacuum. The 7 compact dimensions are necessary to wrap around this 14-dimensional defect.

4 Cosmological Selection: Inflation as Filtration

The *Big Bang* is identified as the phase transition $\mathbb{T} \rightarrow \mathbb{S} \rightarrow \mathbb{O}$.

4.1 Inflationary Dilution

The density of zero divisors in the pre-geometric bulk scales exponentially with dimension. To stabilize the Octonionic sector ($\mathbb{O}$), the vacuum initiates **Cosmic Inflation**.

$$a(t) \propto e^{Ht} \quad (6)$$

This exponential expansion is not a creation event but a **Dilution Mechanism**. It drives the local density of zero-divisor regions $\rho_{\mathcal{Z}}$ to zero, pushing the algebraic singularities beyond the causal horizon. Inflation halts when the vacuum crystallizes into the division algebra $\mathbb{O}$, where local spatial interactions become reversible ($xy = 0 \implies x = 0$).

4.2 Cosmic Voids as Fossils

Regions of the vacuum that failed to clear the zero-divisor condition during the phase transition persist as **Cosmic Supervoids**.

$$\text{Void} \equiv \text{Region where } \langle \mathcal{Z}(\mathbb{S}) \rangle \neq 0 \quad (7)$$

We predict that these voids are not empty but **Causally Disconnected**. They should exhibit anomalous lensing and photon propagation properties (opacity) due to the failure of the metric connection in the zero-divisor limit.

5 Parameter Estimation: The Sedenion Resonance

We utilize the scale-invariant Thin Filament Code (TFC) solver to estimate the vibrational frequency of the vacuum geometry at the Sedenion transition. This solver, originally developed for magnetospheric flux tubes (Wolf et al., 2012), relies on the isomorphism between plasma stability and vacuum geometry derived in Schutza (2025).

5.1 The π -Stiffness Hypothesis

In previous work, we derived the effective stiffness of the QCD (Octonionic) vacuum to be $\beta_{\mathbb{O}} \approx 1.91$, corresponding to a polytropic index $\Gamma_{\mathbb{O}} \approx 2.91$. For the Sedenion transition, we postulate that the vacuum stiffness saturates the geometric limit of the flux tube topology.

$$\Gamma_{\mathbb{S}} \equiv \pi \approx 3.14159 \quad (8)$$

This represents the maximum hoop stress supportable by the geometry before the metric connection fails (i.e., the ratio of circumference to diameter).

5.2 Scaling Prediction

The fundamental eigenfrequency ω_0 in the TFC Sturm-Liouville problem scales with the acoustic velocity, and thus with the square root of the stiffness: $\omega \propto \sqrt{\Gamma}$. Using the QCD glueball mass ($\omega_{QCD} \approx 6.55$) as the baseline:

$$\omega_{\mathbb{S}} = \omega_{QCD} \times \sqrt{\frac{\Gamma_{\mathbb{S}}}{\Gamma_{\mathbb{O}}}} \approx 6.55 \times \sqrt{\frac{\pi}{2.91}} \quad (9)$$

$$\omega_{\mathbb{S}} \approx 6.55 \times 1.039 \approx \mathbf{6.81} \quad (\text{Dimensionless}) \quad (10)$$

This dimensionless frequency represents the “breathing mode” of the Sedenion geometry just before fracture.

6 The Sedenion Scream: A Prediction at 1.8 GHz

To convert the dimensionless resonance into a physical observable, we scale by the characteristic energy of the symmetry breaking.

6.1 Thermal Scaling from the GUT Epoch

We identify the Sedenion-to-Octonion filtration scale with the Grand Unified Theory (GUT) scale, $E_{GUT} \approx 2 \times 10^{16}$ GeV, where the unified gauge couplings diverge. The characteristic frequency at emission is $f_* \sim H_*$. The observed frequency today f_{obs} is redshifted by the expansion of the universe:

$$f_{obs} \approx f_* \left(\frac{T_0}{T_*} \right) \approx 1.65 \times 10^{-7} \text{ Hz} \left(\frac{E_{GUT}}{1 \text{ GeV}} \right) \quad (11)$$

Substituting the TFC resonance factor $\frac{\omega_{\mathbb{S}}}{2\pi} \approx 1.08$:

$$f_{peak} \approx 1.08 \times (1.65 \times 10^{-7}) \times (10^{16}) \text{ Hz} \quad (12)$$

$$f_{peak} \approx 1.78 \times 10^9 \text{ Hz} \approx \mathbf{1.8 \text{ GHz}} \quad (13)$$

Prediction: The phase transition generates a distinct, narrow-band **Blue Peak** in the Stochastic Gravitational Wave Background (SGWB) at 1.8 GHz. We term this signal the “Sedenion Scream.”

6.2 Engineering Feasibility: Detection via Resonant Cavities

The predicted signal at $f_{peak} \approx 1.8$ GHz ($\lambda \approx 16$ cm) lies in the microwave *L*-band, a regime inaccessible to interferometric detectors like LIGO or LISA. However, this frequency is ideal for detection via the **Inverse Gertsenshtein Effect** (graviton-photon conversion).

Detection Mechanism: A gravitational wave of amplitude h propagating through a static magnetic field B_0 induces an oscillating electromagnetic field E_{GW} :

$$E_{GW} \propto h B_0 L Q \quad (14)$$

where L is the interaction length and Q is the quality factor of the cavity. This conversion is resonantly enhanced when the cavity frequency matches the GW frequency.

Experimental Synergy: The required apparatus—a high- Q superconducting cavity in a multi-Tesla magnetic field tuned to GHz frequencies—is identical to the hardware currently operating for Axion Dark Matter searches (e.g., **ADMX**).

- **Target Frequency:** 1.8 GHz aligns directly with the sensitivity range of ADMX Run 1C/1D.
- **Signal Discrimination:** A Sedenion-origin GW signal would appear as a narrow-band power excess. Unlike an axion signal, the GW conversion efficiency depends on the relative orientation of the cavity mode and the tensor polarization of the wave.

7 Conclusion

We have derived the G_2 holonomy of M-theory from the algebraic properties of the Sedenions. The universe exists in a delicate balance: it rides the **Power-Associative Lifeline** of time while constantly filtering out the zero-divisor instabilities of space. The detection of a resonant gravitational wave peak at 1.8 GHz would constitute direct experimental evidence of this pre-geometric filtration process.